Geometry: Similarity in Polygons

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Name: _____ Date: _____ Score: / 10

DIRECTIONS *Determine similarity, find missing values, and justify your answers using angle and side relationships.*

1 Are the polygons similar? State why or why not:

$\triangle ABC: \angle A = 50°, \angle B = 60°, \angle C = 70°$ $\triangle DEF: \angle D = 50°, \angle E = 60°, \angle F = 70°$

Answer: _____

2 Find the scale factor from ABCD to EFGH:

$AB = 6,\ BC = 9,\ EF = 2,\ FG = 3$

Answer: _____

3 Solve for x if the polygons are similar:

$$\frac{x}{4} = \frac{9}{6}$$

Answer: _____

4 Find the missing angle if polygons are similar:

Polygon 1 angles: 80°, 100°, 75°, 105° Polygon 2 angles: 80°, 100°, 75°, ?

Answer: _____

5 Are the rectangles similar? Explain:

Rectangle 1: 4×10 Rectangle 2: 6×15

Answer: _____

6 Solve for x and y if the quadrilaterals are similar:

$$\frac{x}{5} = \frac{12}{10},\quad \frac{y}{8} = \frac{12}{10}$$

Answer: _____

7 Determine if the triangles are similar:

$\triangle PQR: PQ = 8,\ QR = 12,\ PR = 16$ $\triangle XYZ: XY = 6,\ YZ = 9,\ XZ = 12$

Answer: _____

8 Find the perimeter of the larger polygon:

Smaller polygon perimeter: 24 cm, Scale factor: $\frac{5}{2}$

Answer: _____

9 Write a similarity statement for the polygons:

Pentagon $ABCDE$ with $\angle A \cong \angle F,\ \angle B \cong \angle G,\ \angle C \cong \angle H,\ \angle D \cong \angle I,\ \angle E \cong \angle J$ and proportional sides

Answer: _____

10 Solve for x given similar polygons:

$$\frac{2x+3}{7} = \frac{15}{5}$$

Answer: _____

11 Are the polygons similar? Find the scale factor:

Hexagon 1 sides: 3, 6, 9, 12, 6, 9 Hexagon 2 sides: 4, 8, 12, 16, 8, 12 All corresponding angles congruent

Answer: _____

12 Solve for both x and y:

Similar quadrilaterals with sides: $\frac{3x-1}{10} = \frac{10}{25}$ and $\frac{y+2}{6} = \frac{10}{25}$

Answer: _____

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Answer Key & Solutions

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TEACHER NOTES *Remind students that two polygons are similar if and only if all corresponding angles are congruent AND all corresponding sides are proportional. Encourage students to write similarity statements in the correct vertex order to reinforce corresponding parts. Stress that the scale factor is always written as a ratio of the first polygon*

1 Are the polygons similar? State why or why not:

$\triangle ABC: \angle A = 50°,\ \angle B = 60°,\ \angle C = 70°$ $\triangle DEF: \angle D = 50°,\ \angle E = 60°,\ \angle F = 70°$

= $\triangle ABC \sim \triangle DEF$

All three corresponding angles are congruent (50°, 60°, 70°), so the triangles are similar by the Angle-Angle (AA) criterion.

2 Find the scale factor from ABCD to EFGH:

$AB = 6,\ BC = 9,\ EF = 2,\ FG = 3$

= $\frac{AB}{EF} = \frac{6}{2} = 3$

Divide a side of ABCD by the corresponding side of EFGH: 6 ÷ 2 = 3, confirming a scale factor of 3:1.

3 Solve for x if the polygons are similar:

$\frac{x}{4} = \frac{9}{6}$

= $x = 6$

Cross-multiply: 6x = 36, then divide both sides by 6 to get x = 6.

4 Find the missing angle if polygons are similar:

Polygon 1 angles: 80°, 100°, 75°, 105° Polygon 2 angles: 80°, 100°, 75°, ?

= $\angle = 105°$

Corresponding angles of similar polygons are congruent, so the missing angle must equal 105° to match the corresponding angle in Polygon 1.

5 Are the rectangles similar? Explain:

Rectangle 1: 4 × 10 Rectangle 2: 6 × 15

= $\frac{4}{6} = \frac{10}{15} = \frac{2}{3}$

Check the ratios: 4/6 = 2/3 and 10/15 = 2/3. Since corresponding sides are proportional and all angles are 90°, the rectangles are

6 Solve for x and y if the quadrilaterals are similar:

$\frac{x}{5} = \frac{12}{10},\quad \frac{y}{8} = \frac{12}{10}$

= $x = 6,\ y = 9.6$

For x: x = 5 × (12/10) = 6. For y: y = 8 × (12/10) = 9.6.

7 Determine if the triangles are similar:

$\triangle PQR: PQ = 8,\ QR = 12,\ PR = 16$ $\triangle XYZ: XY = 6,\ YZ = 9,\ XZ = 12$

= $\frac{8}{6} = \frac{12}{9} = \frac{16}{12} = \frac{4}{3}$

All three pairs of corresponding sides share the same ratio 4:3, so the triangles are similar by SSS Similarity.

8 Find the perimeter of the larger polygon:

Smaller polygon perimeter: 24 cm, Scale factor: $\frac{5}{2}$

= $P = 24 \cdot \frac{5}{2} = 60 \text{ cm}$

Multiply the smaller perimeter by the scale factor: 24 × (5/2) = 60 cm.

9 Write a similarity statement for the polygons:

Pentagon *ABCDE* with $\angle A \cong \angle F,\ \angle B \cong \angle G,\ \angle C \cong \angle H,\ \angle D \cong \angle I,\ \angle E \cong \angle J$ and proportional sides

= $ABCDE \sim FGHIJ$

Since all corresponding angles are congruent and sides are proportional, write the similarity statement matching vertices in order: ABCDE ~ FGHIJ.

10 Solve for x given similar polygons:

$\frac{2x+3}{7} = \frac{15}{5}$

= $x = 9$

Cross-multiply: 5(2x + 3) = 105, so 10x + 15 = 105, giving 10x = 90 and x = 9.

11 Are the polygons similar? Find the scale factor:

Hexagon 1 sides: 3, 6, 9, 12, 6, 9 Hexagon 2 sides: 4, 8, 12, 16, 8, 12 All corresponding angles congruent

= $\frac{3}{4} = \frac{6}{8} = \frac{9}{12} = \frac{12}{16} = \frac{3}{4}$

All corresponding side ratios equal 3/4 and all angles are congruent, so the hexagons are similar with a scale factor of 3:4.

12 Solve for both x and y:

Similar quadrilaterals with sides: $\frac{3x-1}{10} = \frac{10}{25}$ and $\frac{y+2}{6} = \frac{10}{25}$

= $x = \frac{5}{3},\ y = \frac{2}{5}$

For x: 3x – 1 = 10 × (10/25) = 4, so 3x = 5 and x = 5/3. For y: y + 2 = 6 × (10/25) = 2.4, so y = 0.4 = 2/5.

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